

# GalaxEye — Take-Home Assignment

## Backend Engineer, ML Systems

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### Context

At GalaxEye, much of our work is turning multi-sensor satellite imagery into usable intelligence — and a good deal of it has to run **offline, on isolated hardware, with no internet**. This exercise is a small, deliberately open-ended slice of that world.

### The scenario

Imagine a service that ingests satellite image tiles, classifies each one by land-use type (forest, water, residential, and so on) using a locally-run model, stores the results, and lets an analyst query them — entirely offline.

We provide you with the satellite tile data: *Galaxeye-BE\_MLSys-TakeHome\_Assignment-Tiles.zip*

### What we'd like from you

*There are three parts.*

#### Part 1 · Your approach (the main event)

Before writing much code, write a short **design note** (1–2 pages, informal is fine) covering:

- **How you'd build the whole thing** — the components, how a tile flows through the system, and how you'd store results so an analyst can query them.
- **The real decisions and trade-offs** you're weighing — for example, how to handle predictions the model isn't confident about, what to store, and what "querying the results" should mean. Tell us the options you considered and how you'd choose between them.
- **Your assumptions, and the questions you'd ask us** if you could. Spotting what's ambiguous is part of the job.

#### Part 2 · A working slice (prove it runs)

Implement the **core path**: one endpoint that takes a tile, runs a classifier, and stores the result. We want evidence you can turn a design into working code — not a finished product. Stub or skip the rest, and say so.

#### Part 3 · Problem-solving (answer in your own words)

*Short answers are fine. We're looking for how you reason, not how much you write.*

1. Your classifier turns out to be **wrong about 30% of the time**. What do you do — and how would you even decide whether it's "good enough" to be useful at all?
2. This service runs **offline, with no one watching it live**. A month after deployment, how would you know it's still working correctly?
3. Tiles are coming in fine, but the **stored results look wrong**. Walk us through how you'd find the cause — your actual steps, in order.
4. What's the **weakest part of your design**, and what would break it first?

## Ground rules

- **It must run offline.** No hosted model APIs (OpenAI, Claude, cloud vision, etc.). Use any locally runnable, pre-trained model you can run on CPU.
- **You can train a model if you wish, but we are fine with a pretrained model also, and we are not grading accuracy.** A model that's often wrong is completely fine — what matters is how you reason it.
- **Use whatever stack you know best.**
- **You may use AI tools** (Copilot, ChatGPT, Claude, and so on). We'd rather you use them well than pretend you didn't. But you must fully understand, and be able to explain and modify, everything you submit — we'll go through it together afterwards.

## What to submit

- Your **design note** (Part 1).
- The **thin slice** of code, with a short README on how to run it (Part 2).
- Your **Part 3 answers** (a plain text file or a section in the README is fine).

How to submit: [Take Home Assignment Submission Form](#)

## What happens next

We'll set up a **interview** to walk through your solution together — you explaining your choices, and a small change or two made live.

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*We're looking forward to seeing how you approach this. Good luck!*